

JEE-MAINS-2020-JAN-9-SHIFT-1

PHYSICS

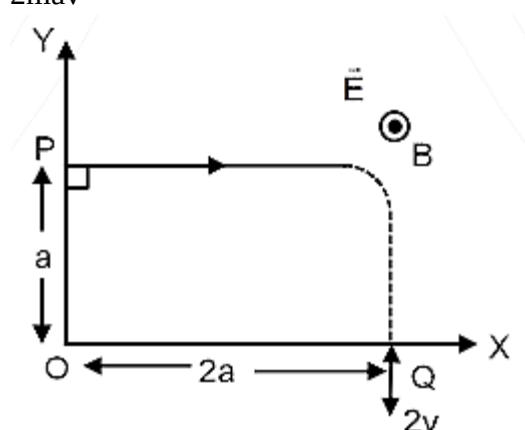
Question 1: A charged particle of mass 'm' and charge 'q' moving under the influence of uniform electric field $\vec{E}$ and a uniform magnetic field $\vec{B}$ follows a trajectory from point P to Q as shown in figure. The velocities at P and Q are respectively, $\vec{v}_i$ and $-2\vec{v}_j$. Then which of the following statements (A, B, C, D) are the correct ? (Trajectory shown is schematic and not to scale)

(A) $E = \frac{3}{4} \left(\frac{mv^2}{qa} \right)$

(B) Rate of work done by the electric field at P is $\frac{3}{4} \left(\frac{mv^3}{a} \right)$

(C) Rate of work done by both the fields at Q is zero

(D) The difference between the magnitude of angular momentum of the particle at P and Q is $2mav$



Options:

- (a) (A), (B), (C)
- (b) (A), (B), (C), (D)
- (c) (A), (C), (D)
- (d) (B), (C), (D)

Question 2: A vessel of depth $2h$ is half filled with a liquid of refractive index $2\sqrt{2}$ and the upper half with another liquid of refractive index $\sqrt{2}$. The liquids are immiscible. The apparent depth of the inner surface of the bottom of vessel will be

Options:

- (a) $\frac{h}{3\sqrt{2}}$
- (b) $\frac{3}{4}h\sqrt{2}$
- (c) $\frac{h}{\sqrt{2}}$

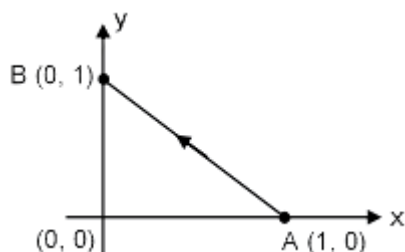
(d) $\frac{h}{2(\sqrt{2}+1)}$

Question 3: The aperture diameter of a telescope is 5m. The separation between the moon and the earth is 4×10^5 km. With light of wavelength of $5500 \text{ \AA}$, the minimum separation between objects on the surface of moon, so that they are just resolved, is close to

Options:

- (a) 600 m
- (b) 20 m
- (c) 200 m
- (d) 60m

Question 4: Consider a force $\vec{F} = -x\hat{i} + y\hat{j}$. The work done by this force in moving a particle from point A(1, 0) to B(0, 1) along the line segment is



Options:

- (a) $\frac{1}{2}$
- (b) $\frac{3}{2}$
- (c) 2
- (d) 1

Question 5: Three harmonic waves having equal frequency ν and same intensity I_0 , have phase angles 0 , $\frac{\pi}{4}$ and $-\frac{\pi}{4}$ respectively. When they are superimposed the intensity of the resultant wave is close to

Options:

- (a) $5.8 I_0$
- (b) $3 I_0$
- (c) I_0
- (d) $0.2 I_0$

Question 6: A quantity f is given by $\sqrt{\frac{hc^5}{G}}$ where c is speed of light, G universal

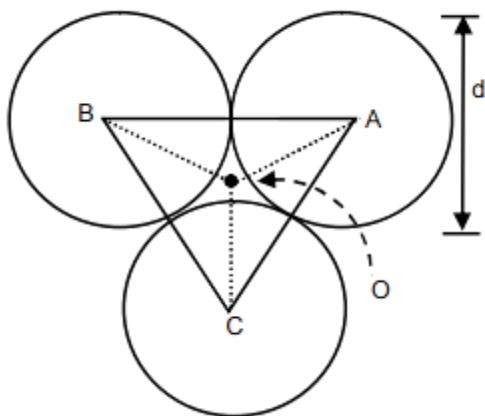
gravitational constant and h is the Planck's constant. Dimension of f is that of

Options:

- (a) Volume

- (b) Energy
- (c) Momentum
- (d) Area

Question 7: Three solid spheres each of mass m and diameter d are stuck together such that the lines connecting the centres form an equilateral triangle of side of length d . The ratio I_0/I_A of moment of inertia I_0 of the system about an axis passing the centroid and about center of any of the spheres A and perpendicular to the plane of the triangle is



Options:

- (a) $\frac{23}{13}$
- (b) $\frac{13}{15}$
- (c) $\frac{13}{23}$
- (d) $\frac{15}{13}$

Question 8: An electric dipole of moment $\vec{p} = (-\hat{i} - 3\hat{j} + 2\hat{k}) \times 10^{-29}$ C.m. is at the origin (0, 0, 0). The electric field due to this dipole at $\vec{r} = (+\hat{i} + 3\hat{k} + 5\hat{k})$ (note that $\vec{r} \cdot \vec{p} = 0$) is parallel to

Options:

- (a) $(+\hat{i} + 3\hat{j} - 2\hat{k})$
- (b) $(-\hat{i} + 3\hat{j} - 2\hat{k})$
- (c) $(+\hat{i} - 3\hat{j} - 2\hat{k})$
- (d) $(-\hat{i} - 3\hat{j} + 2\hat{k})$

Question 9: Radiation, with wavelength 6561 Å falls on a metal surface to produce photoelectrons. The electrons are made to enter a uniform magnetic field of 3×10^{-4} T. If the radius of the largest circular path followed by the electrons is 10 mm, the work function of the metal is close to

Options:

- (a) 1.8 eV
- (b) 0.8 eV
- (c) 1.6 eV
- (d) 1.1 eV

Question 10: A long, straight wire of radius a carries a current distributed uniformly over its cross-section. The ratio of the magnetic fields due to the wire at distance $\frac{a}{3}$ and $2a$, respectively from the axis of the wire is

Options:

- (a) $\frac{1}{2}$
- (b) 2
- (c) $\frac{3}{2}$
- (d) $\frac{2}{3}$

Question 11: Consider a sphere of radius R which carries a uniform charge density ρ . If a sphere of radius $\frac{R}{2}$ is carved out of it, as shown, the ratio $\left| \frac{\vec{E}_A}{\vec{E}_B} \right|$ of magnitude of electric field $\vec{E}_A$ and $\vec{E}_B$ respectively, at points A and B due to the remaining portion is

Options:

- (a) $\frac{18}{34}$
- (b) $\frac{21}{34}$
- (c) $\frac{17}{54}$
- (d) $\frac{18}{54}$

Question 12: A body A of mass m is moving in a circular orbit of radius R about a planet.

Another body B of mass $\frac{m}{2}$ collides with A with a velocity which is half $\left(\frac{\vec{v}}{2} \right)$ the

instantaneous velocity $\vec{v}$ of A. The collision is completely inelastic. Then, the combined body

Options:

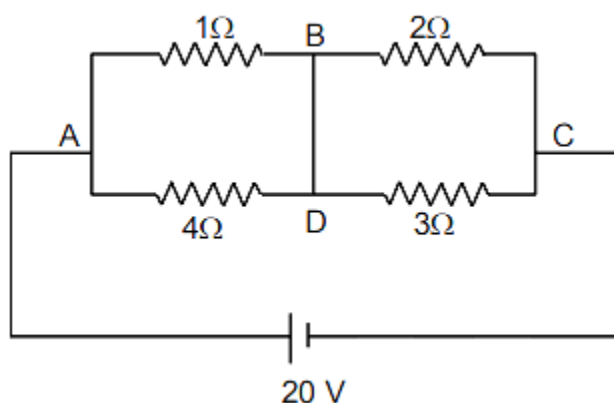
- (a) Escapes from the planet's Gravitational field
- (b) Continues to move in a circular orbit
- (c) Falls vertically downwards towards the planet
- (d) Starts moving in a elliptical orbit around the planet

Question 13: If the screw on a screw-gauge is given six rotations, it moves by 3mm on the main scale. If there are 50 divisions on the circular scale the least count of the screw gauge is

Options:

- (a) 0.01 cm
- (b) 0.02 mm
- (c) 0.001 mm
- (d) 0.001 cm

Question 14: In the given circuit diagram, a wire is joining points B and D. The current in this wire is



Options:

- (a) Zero
- (b) 2A
- (c) 0.4 A
- (d) 4A

Question 15: The electric fields of two plane electromagnetic plane waves in vacuum are given $\vec{E}_1 = E_0 \hat{j} \cos(\omega t - kx)$ and $\vec{E}_2 = \hat{k} \cos(\omega t - ky)$. At $t = 0$, a particle of charge q is at origin with a velocity $\vec{v} = 0.8c\hat{j}$ (c is the speed of the light in vacuum). The instantaneous force experienced by the particle is

Options:

- (a) $E_0 q (0.8\hat{i} + \hat{j} + 0.2\hat{k})$
- (b) $E_0 q (-0.8\hat{i} + \hat{j} + \hat{k})$
- (c) $E_0 q (0.4\hat{i} + 3\hat{j} + 0.8\hat{k})$
- (d) $E_0 q (0.8\hat{i} - \hat{j} + 0.4\hat{k})$

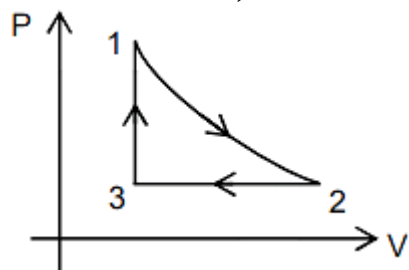
Question 16: Consider two ideal diatomic gases A and B at some temperature T . Molecules of the gas A are rigid, and have a mass m . Molecules of the gas B have an additional vibrational mode, and have a mass $\frac{m}{4}$. The ratio of the specific heats (C_V^A and C_V^B) of gas A and B, respectively is

Options:

- (a) 3 : 5
- (b) 5 : 9
- (c) 7 : 9

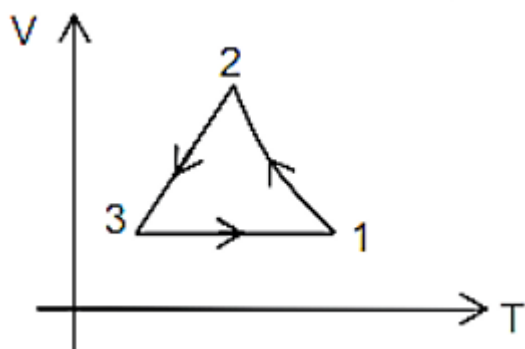
(d) 5 : 7

Question 17: Which of the following is an equivalent cyclic process corresponding to the thermodynamic cyclic given in the figure ? where, $1 \rightarrow 2$ is adiabatic. (Graphs are schematic and are not to scale)

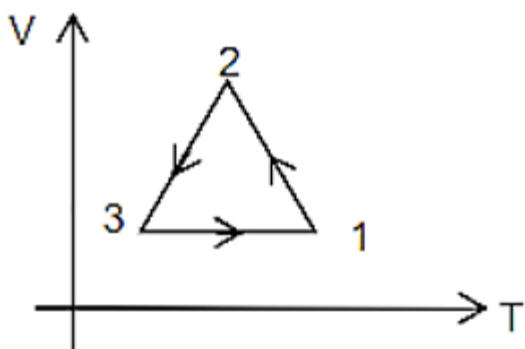


Options:

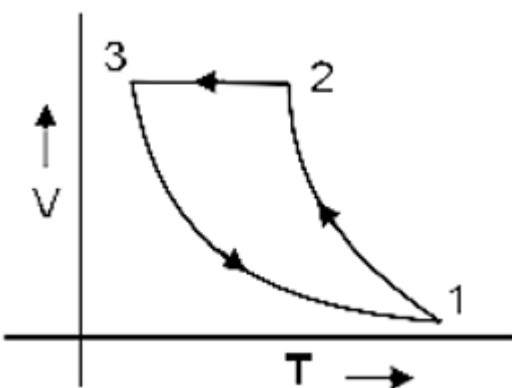
(a)



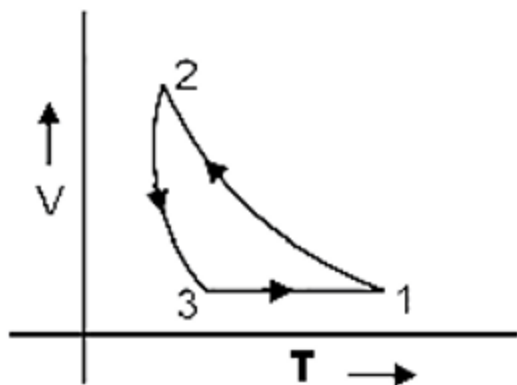
(b)



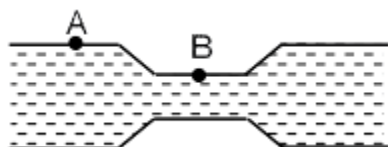
(c)



(d)



Question 18: Water flows in a horizontal tube (see figure). The pressure of water changes by 700 Nm^{-2} between A and B where the area of cross section are 40 cm^2 and 20 cm^2 , respectively. Find the rate of flow of water through the tube.
(density of water = 1000 kgm^{-3})



Options:

- (a) $2720 \text{ cm}^3/\text{s}$
- (b) $2420 \text{ cm}^3/\text{s}$
- (c) $3020 \text{ cm}^3/\text{s}$
- (d) $1810 \text{ cm}^3/\text{s}$

Question 19: Two particles of equal mass m have respective initial velocities $u\hat{i}$ and $u\left(\frac{\hat{i} + \hat{j}}{2}\right)$. They collide completely inelastically. The energy lost in the process is

Options:

- (a) $\frac{3}{4}mu^2$
- (b) $\sqrt{\frac{2}{3}}mu^2$
- (c) $\frac{1}{3}mu^2$
- (d) $\frac{1}{8}mu^2$

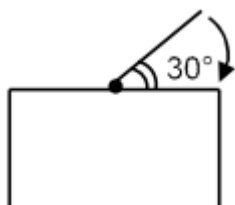
Question 20: A particle moving with kinetic energy E has de Broglie wavelength λ . If energy ΔE is added to its energy, the wavelength become $\lambda/2$. Value of ΔE , is

Options:

- (a) $4E$
- (b) $3E$

- (c) $2E$
(d) E

Question 21: One end of a straight uniform 1 m long bar is pivoted on horizontal table. It is released from rest when it makes an angle 30° from the horizontal (see figure). Its angular speed when it hits the table is given as



Question 22: The distance x covered by a particle in one dimensional motion varies with time t as $x^2 = at^2 + 2bt + c$. If the acceleration of the particle depends on x as x^{-n} , where n is an integer, the value of n is

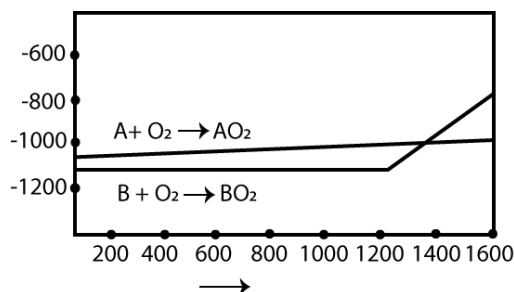
Question 23: A body of mass $m = 10$ kg is attached to one end of a wire of length 0.3 m. The maximum angular speed (in rad s^{-1}) with which it can be rotated about its other end in space station is (Breaking stress of wire $= 4.8 \times 10^7 \text{ Nm}^{-2}$ and area of cross section of the wire $= 10^{-2} \text{ cm}^2$) is

Question 24: Both the diodes used in the circuit shown are assumed to be ideal and have negligible resistance when these are forward biased. built in potential in each diode is 0.7 V. For the input voltages shown in the figure, the voltage (in volts) at point A is

Question 25: In a fluorescent lamp choke (a small transformer) 100 V of reverse voltage is produced when the choke current changes uniformly from 0.25 A to 0 in a duration of 0.025 ms. The self-inductance of the choke (in mH) is estimated to be

CHEMISTRY

Question 26: According to the following diagram, A reduces BO_2 when the temperature is :



Options:

- (a) $< 1200^\circ\text{C}$
(b) $> 1200^\circ\text{C}$ but $< 1400^\circ\text{C}$
(c) $< 1400^\circ\text{C}$
(d) $> 1400^\circ\text{C}$

Question 27: The acidic, basic and amphoteric oxides, respectively, are :

Options:

- (a) Na_2O , SO_3 , Al_2O_3
- (b) N_2O_3 , Li_2O , Al_2O_3
- (c) Cl_2O , CaO , P_4O_{10}
- (d) MgO , Cl_2O , Al_2O_3

Question 28: If the magnetic moment of a dioxygen species is 1.73 B.M., it may be :

Options:

- (a) O_2^- or O_2^+
- (b) O_2 or O_2^+
- (c) O_2 or O_2^-
- (d) O_2 , O_2^- or O_2^+

Question 29: The K_{sp} for the following dissociation is 1.6×10^{-5}

$\text{PbCl}_2(s) \rightleftharpoons \text{Pb}_{(aq)}^{2+} + 2\text{Cl}_{(aq)}^-$ Which of the following choices is correct for a mixture of 300 mL 0.134M $\text{Pb}(\text{NO}_3)_2$ and 100 mL 0.4 M NaCl ?

Options:

- (a) $Q = K_{sp}$
- (b) $Q > K_{sp}$
- (c) Not enough data provided
- (d) $Q < K_{sp}$

Question 30: If enthalpy of atomisation for $\text{Br}_2(\text{l})$ is x kJ/mol and bond enthalpy for Br_2 is y kJ/mol the relation between them :

Options:

- (a) is $x = y$
- (b) is $x < y$
- (c) does not exist
- (d) is $x > y$

Question 31: $[\text{Pd}(\text{F})(\text{Cl})(\text{Br})(\text{I})]^{2-}$ has n number of geometrical isomers. Then, the spin-only magnetic moment and crystal field stabilisation energy [CFSE] of $[\text{Fe}(\text{CN})_6]^{n-6}$, respectively, are : [Note : Ignore the pairing energy]

Options:

- (a) 5.92 BM and 0
- (b) 0 BM and $-2.4 \Delta_0$
- (c) 1.73 BM and $-2.0 \Delta_0$
- (d) 2.84 BM and $-1.6 \Delta_0$

Question 32: The de Broglie wavelength of an electron in the 4th Bohr orbit is

Options:

- (a) $4\pi a_0$
- (b) $2\pi a_0$

- (c) $6\pi a_0$
(d) $8\pi a_0$

Question 33: The compound that cannot act both as oxidising and reducing agent is :

Options:

- (a) HNO_2
(b) H_3PO_4
(c) H_2SO_3
(d) H_2O_2

Question 34: B has a smaller first ionization enthalpy than Be. Consider the following statements :

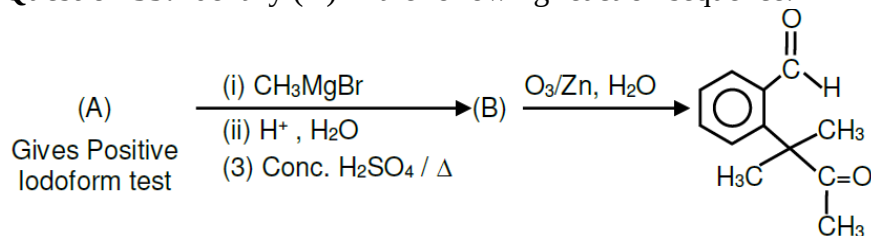
- (I) It is easier to remove 2p electron than 2s electron
(II) 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be
(III) 2s electron has more penetration power than 2p electron
(IV) atomic radius of B is more than Be (atomic number B = 5, Be = 4)

The correct statement are :

Options:

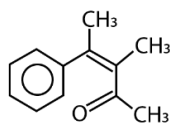
- (a) (I), (III) and (IV)
(b) (II), (III) and (IV)
(c) (I), (II) and (IV)
(d) (I), (II) and (III)

Question 35: Identify (A) in the following reaction sequence:

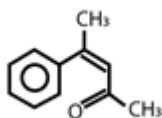


Options:

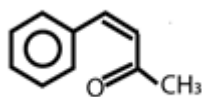
(a)



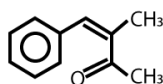
(b)



(c)



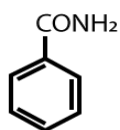
(d)



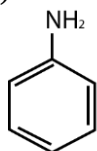
Question 36: Which of these will produce the highest yield in Friedel Craft reaction ?

Options:

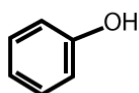
(a)



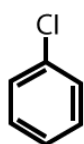
(b)



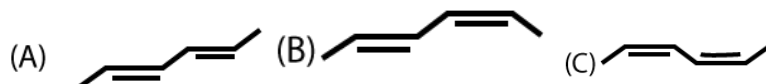
(c)



(d)



Question 37: The correct order of heat of combustion for following alkadienes is :



Options:

(a) (B) < (C) < (A)

(b) (C) < (B) < (A)

(c) (A) < (B) < (C)

(d) (A) < (C) < (B)

Question 38: 'X' melts at low temperature and is a bad conductor of electricity in both liquid and solid state. X is :

Options:

- (a) Carbon tetrachloride
- (b) Zinc sulphide
- (c) Silicon carbide
- (d) Mercury

Question 39: A chemist has 4 samples of artificial sweetener A, B, C and D. To identify these samples, he performed certain experiments and noted the following observations :

- (i) A and D both form blue-violet colour with ninhydrin.
- (ii) Lassaigne extract of C gives positive AgNO_3 test and negative $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ test.
- (iii) Lassaigne extract of B and D gives positive sodium nitroprusside test.

Based on these observations which option is correct ?

Options:

- (a) A : Alitame ; B : Saccharin ; C : Aspartame; D : Sucralose
- (b) A : Aspartame; B : Alitame ; C : Saccharin ; D : Sucralose
- (c) A : Aspartame; B : Saccharin ; C : Sucralose ; D : Alitame
- (d) A : Saccharin ; B : Alitame ; C : Sucralose ; D : Aspartame

Question 40: The electronic configurations of bivalent europium and trivalent cerium are :
(atomic number : Xe = 54, Ce = 58, Eu = 63)

Options:

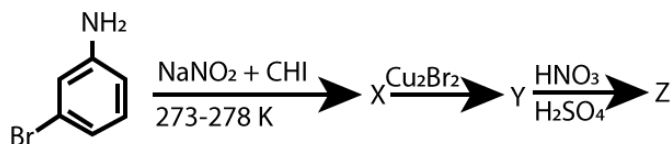
- (a) $[\text{Xe}] 4f^4$ and $[\text{Xe}] 4f^9$
- (b) $[\text{Xe}] 4f^7$ and $[\text{Xe}] 4f^1$
- (c) $[\text{Xe}] 4f^7 6s^2$ and $[\text{Xe}] 4f^2 6s^2$
- (d) $[\text{Xe}] 4f^2$ and $[\text{Xe}] 4f^7$

Question 41: Complex X of composition of $\text{Cr}(\text{H}_2\text{O})_6\text{Cl}_n$ has a spin only magnetic moment of 3.83 BM. It reacts with AgNO_3 and shows geometrical isomerism. The IUPAC nomenclature of X is :

Options:

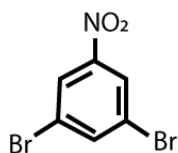
- (a) Dichloridotetraaqua chromium(IV) chloride dehydrate
- (b) Hexaaqua chromium(III)chloride
- (c) Tetraaquadichlorido chromium(III) chloride dehydrate
- (d) Tetraaquadichlorido chromium(IV) chloride dihydrate

Question 42: The major product Z obtained in the following reaction scheme is

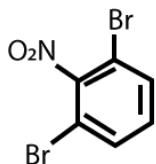


Options:

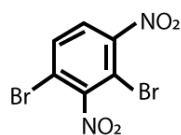
- (a)



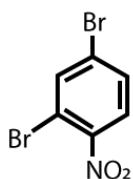
(b)



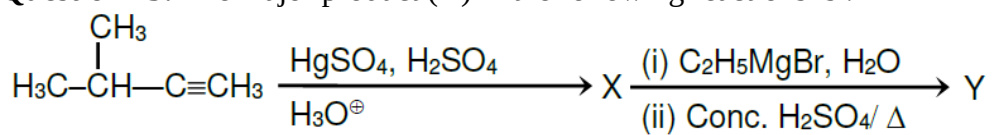
(c)



(d)

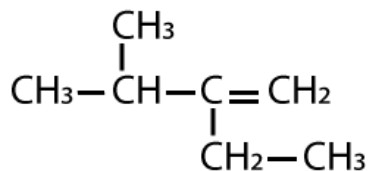


Question 43: The major product (Y) in the following reactions is :

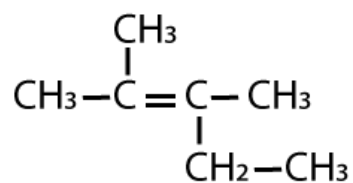


Options:

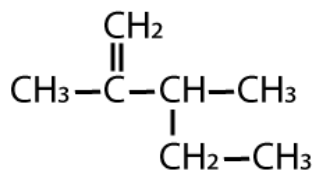
(a)



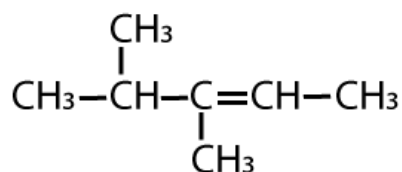
(b)



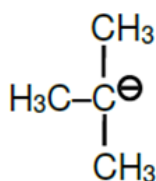
(c)



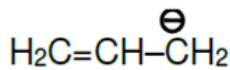
(d)



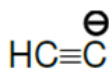
Question 44: The increasing order of basicity for the following intermediates is (from weak to strong)



(i)



(ii)



(iii)



(iv)



(v)

Options:

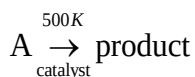
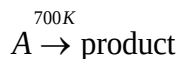
(a) (iii) < (iv) < (ii) < (i) < (v)

(b) (iii) < (i) < (ii) < (iv) < (v)

(c) (v) < (iii) < (ii) < (iv) < (i)

(d) (v) < (i) < (iv) < (ii) < (iii)

Question 45: For following reactions



it was found that the E_a is decreased by 30 kJ/mol in the presence of catalyst. If the rate remains unchanged, the activation energy for catalysed reaction is (Assume pre exponential factor is same):

Options:

(a) 105 kJ/mol

(b) 195 kJ/mol

- (c) 135 kJ/mol
(d) 75 kJ/mol

Question 46: 108 g of silver (molar mass 108 g mol^{-1}) is deposited at cathode from $\text{AgNO}_3(\text{aq})$ solution by a certain quantity of electricity. The volume (in L) of oxygen gas produced at 273 K and 1 bar pressure from water by the same quantity of electricity is _____

Question 47: How much amount of NaCl should be added to 600 g for water ($\rho = 1.00 \text{ g/mL}$) to decrease the freezing point of water to -0.2°C ? _____. (The freezing point depression constant of water = 2 K kg mol^{-1}).

Question 48: The hardness of a water sample containing 10^{-3} M MgSO_4 expressed as CaCO_3 equivalents (in ppm) is _____.

Question 49: The mass percentage of nitrogen in histamine is _____

Question 50: The molarity of HNO_3 in a sample which has density 1.4 g/mL and mass percentage of 63% is _____. (Molecular Weight of $\text{HNO}_3 = 63$)

MATHEMATICS

Question 51: The product $2^{\frac{1}{4}} \cdot 4^{\frac{1}{16}} \cdot 8^{\frac{1}{48}} \cdot 16^{\frac{1}{128}} \dots$ to ∞ is equal to

Options:

- (a) $2^{\frac{1}{4}}$
(b) $2^{\frac{1}{2}}$
(c) 1
(d) 2

Question 52: If $f'(x) = \tan^{-1}(\sec x + \tan x)$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$, and $f(0) = 0$, then $f(1)$ is equal to :

Options:

- (a) $\frac{1}{4}$
(b) $\frac{\pi + 2}{4}$
(c) $\frac{\pi - 1}{4}$
(d) $\frac{\pi + 1}{4}$

Question 53: The value of $\int_0^{2\pi} \frac{x \sin^8 x}{\sin^8 x + \cos^8 x} dx$ is equal to

Options:

- (a) 2π
- (b) π^2
- (c) 4π
- (d) $2\pi^2$

Question 54: The value of $\cos^3\left(\frac{\pi}{8}\right) \cdot \cos\left(\frac{3\pi}{8}\right) + \sin^3\left(\frac{\pi}{8}\right) \cdot \sin\left(\frac{3\pi}{8}\right)$ is

Options:

- (a) $1/2\sqrt{2}$
- (b) $1/\sqrt{2}$
- (c) $1/4$
- (d) $1/2$

Question 55: If the matrices $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{bmatrix}$, $B = \text{adj } A$ and $C = 3A$, then $\frac{|\text{adj } B|}{|C|} =$

Options:

- (a) 72
- (b) 8
- (c) 16
- (d) 2

Question 56: A circle touches the y-axis at the point (0, 4) and passes through the point (2, 0). Which of the following lines is not a tangent to this circle ?

Options:

- (a) $4x - 3y + 17 = 0$
- (b) $3x - 4y - 24 = 0$
- (c) $3x + 4y - 6 = 0$
- (d) $4x + 3y - 8 = 0$

Question 57: A spherical iron ball of 10 cm radius is coated with a layer of ice of uniform thickness that melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of ice is 5 cm, then the rate (in cm/min.) at which of the thickness of ice decreases, is :

Options:

- (a) $5/6\pi$
- (b) $1/36\pi$
- (c) $1/18\pi$
- (d) $1/54\pi$

Question 58: Negation of the statement : $\sqrt{5}$ is an integer or 5 is irrational is :

Options:

- (a) $\sqrt{5}$ is not an integer or 5 is not irrational

- (b) $\sqrt{5}$ is an integer and 5 is irrational
 (c) $\sqrt{5}$ is irrational or 5 is an integer.
 (d) $\sqrt{5}$ is not an integer and 5 is not irrational

Question 59: The integral $\int \frac{dx}{(x+4)^{8/7} (x-3)^{6/7}}$ is equal to :

Options:

- (a) $\left(\frac{x-3}{x+4}\right)^{1/7} + C$
 (b) $-\frac{1}{13} \left(\frac{x-3}{x+4}\right)^{-13/7} + C$
 (c) $\frac{1}{2} \left(\frac{x-3}{x+4}\right)^{3/7} + C$
 (d) $-\left(\frac{x-3}{x+4}\right)^{-1/7} + C$

Question 60: If the number of five digit numbers with distinct digits and 2 at the 10th place is 336 k, then k is equal to :

Options:

- (a) 7
 (b) 4
 (c) 6
 (d) 8

Question 61: In an box, there are 20 cards, out of which 10 are labelled as A and the remaining 10 are labelled as B. Cards are drawn at random, one after the other and with replacement, till a second A-card is obtained.

The probability that the second A-card appears before the third B-card is :

Options:

- (a) 13/16
 (b) 15/16
 (c) 9/16
 (d) 11/16

Question 62: Let z be a complex number such that $\left|\frac{z-1}{z+2i}\right| = 1$ and $|z| = \frac{5}{2}$. Then the value of

$|z + 3i|$ is :

Options:

- (a) $\sqrt{10}$
 (b) $2\sqrt{3}$
 (c) 7/2
 (d) 15/4

Question 63: If for some α and β in $\mathbb{R}$, the intersection of the following three planes

$$x + 4y - 2z = 1$$

$$x + 7y - 5z = \beta$$

$$x + 5y + z = 5$$

is a line in $\mathbb{R}^3$, then $\alpha + \beta$ is equal to:

Options:

(a) 0

(b) -10

(c) 10

(d) 2

Question 64: The number of real roots of the equation, $e^{4x} + e^{3x} - 4e^{2x} + e^x + 1 = 0$ is

Options:

(a) 3

(b) 1

(c) 4

(d) 2

Question 65: Let f be any function continuous on $[a, b]$ and twice differentiable on (a, b) . If

for all $x \in (a, b)$, $f'(x) > 0$ and $f''(x) < 0$, then for any $c \in (a, b)$, $\frac{f(c) - f(a)}{f(b) - f(c)}$ is greater than

Options:

(a) 1

(b) $\frac{b-c}{c-a}$

(c) $\frac{c-a}{b-c}$

(d) $\frac{b+a}{b-a}$

Question 66: If for all real triplets (a, b, c) , $f(x) = a + bx + cx^2$; then $\int_0^1 f(x) dx =$

Options:

(a) $\frac{1}{3} \left\{ f(0) + f\left(\frac{1}{2}\right) \right\}$

(b) $\frac{1}{3} \left\{ f(1) + 3f\left(\frac{1}{2}\right) \right\}$

(c) $\frac{1}{6} \left\{ f(0) + f(1) + 4f\left(\frac{1}{2}\right) \right\}$

(d) $2 \left\{ 3f(1) + 2f\left(\frac{1}{2}\right) \right\}$

Question 67: Let the observations $x_i (1 \leq i \leq 10)$ satisfy the equations,

$$\sum_{i=1}^{10} (x_i - 5) = 10 \text{ and } \sum_{i=1}^{10} (x_i - 5)^2 = 40. \text{ If } \mu \text{ and } \lambda \text{ are the mean and the variance of the}$$

observations, $x_1 - 3, x_2 - 3, \dots, x_{10} - 3$, then the ordered pair (μ, λ) is equal to

Options:

- (a) (3, 3)
- (b) (3, 6)
- (c) (6, 6)
- (d) (6, 3)

Question 68: If e_1 and e_2 are the eccentricities of the ellipse, $\frac{x^2}{18} + \frac{y^2}{4} = 1$ and the hyperbola,

$$\frac{x^2}{9} - \frac{y^2}{4} = 1 \text{ respectively and } (e_1, e_2) \text{ is a point on the ellipse, } 15x^2 + 3y^2 = k, \text{ then } k \text{ is equal}$$

to

Options:

- (a) 14
- (b) 15
- (c) 16
- (d) 17

Question 69: Let C be the centroid of the triangle with vertices $(3, -1)$, $(1, 3)$ and $(2, 4)$. Let P be the point of intersection of the lines $x + 3y - 1 = 0$ and $3x - y + 1 = 0$. Then the line passing through the points C and P also passes through the point

Options:

- (a) (7, 6)
- (b) $(-9, -7)$
- (c) (9, 7)
- (d) $(-9, -6)$

Question 70: If $f(x) = \begin{cases} \frac{\sin(a+2)x + \sin x}{x}, & x < 0 \\ b, & x = 0 \\ \frac{(x+3x^2)^{\frac{1}{3}} - x^{\frac{1}{3}}}{x^{\frac{4}{3}}}, & x > 0 \end{cases}$ is continuous at $x = 0$, then $a + 2b$ is

equal to :

Options:

- (a) 1
- (b) 0
- (c) -2
- (d) -1

Question 71: The number of distinct solutions of the equation, $\log_{1/2}|\sin x| = 2 - \log_{1/2}|\cos x|$ in the interval $[0, 2\pi]$, is

Question 72: The projection of the line segment joining the points $(1, -1, 3)$ and $(2, -4, 11)$ on the line joining the points $(-1, 2, 3)$ and $(3, -2, 10)$ is :

Question 73: The coefficient of x^4 in the expansion of $(1 + x + x^2)^{10}$ is...

Question 74: If for $x \geq 0$, $y = y(x)$ is the solution of the differential equation, $(x + 1)dy = \{(x + 1)^2 + y - 3\}dx$, $y(2) = 0$, then $y(3)$ is equal to :

Question 75: If the vectors $\vec{p} = (a+1)\hat{i} + a\hat{j} + a\hat{k}$, $\vec{q} = a\hat{i} + (a+1)\hat{j} + a\hat{k}$ and $\vec{r} = a\hat{i} + a\hat{j} + (a+1)\hat{k}$, $(a \in \mathbb{R})$ are coplanar and $3(\vec{p} \cdot \vec{q})^2 - \lambda|\vec{r} \times \vec{q}| = 0$, then the value of λ is _____.

Answer Key:

1. (d)	2. (b)	3. (d)	4. (b)	5. (a)	6. (a)	7. (c)	8. (a)	9. (d)	10. (a)
11. (b)	12. (d)	13. (a)	14. (b)	15. (d)	16. (a)	17. (b)	18. (d)	19. (c)	20. (d)
21. (12.00)	22. (4.00)	23. (3.00)	24. (15.00)	25. (10.00)	26. (d)	27. (b)	28. (a)	29. (b)	30. (d)
31. (c)	32. (d)	33. (b)	34. (d)	35. (d)	36. (d)	37. (c)	38. (a)	39. (c)	40. (b)
41. (c)	42. (d)	43. (b)	44. (c)	45. (d)	46. (5.66)	47. (1.74)	48. (100.00)	49. (37.80)	50. (14.00)
51. (b)	52. (d)	53. (b)	54. (a)	55. (b)	56. (d)	57. (c)	58. (d)	59. (a)	60. (d)
61. (d)	62. (c)	63. (c)	64. (b)	65. (c)	66. (c)	67. (a)	68. (c)	69. (d)	70. (b)
71. (8.00)	72. (8.00)	73. (615.00)	74. (3.00)	75. (1.00)					

SOLUTIONS:

PHYSICS

Solution 1:

(a) According to work energy theorem

$$W_{mag} + W_{ele} = \frac{1}{2} m (2v)^2 - \frac{1}{2} m (v)^2$$

$$0 + qE_0 2a = \frac{3}{2} mv^2$$

$$E_0 = \frac{3}{4} \frac{mv^2}{qa}$$

(b) Rate at which work is done at A should be equivalent to power of electric force

$$A = qE_0 V$$

$$= \frac{3}{4} \frac{mv^3}{a}$$

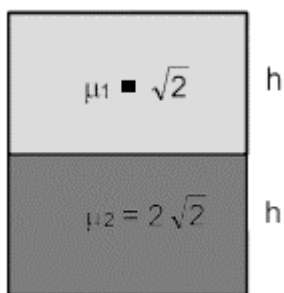
(c) at Q, $\frac{dW}{dt} = 0$ for both forces

$$Q, \frac{dW}{dt} = 0,$$

$$\Delta \vec{L} = (-m2va\vec{k}) - (-mva\hat{k})$$

$$(d) |\Delta \vec{L}| = 3mva$$

Solution 2:

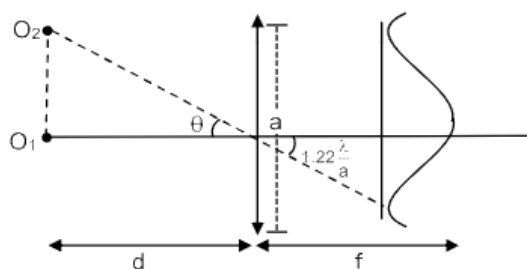


From the figure, we have;

$$d = \frac{h}{\sqrt{2}} + \frac{h}{2\sqrt{2}}$$

$$\Rightarrow d = \frac{h}{\sqrt{2}} \times \frac{3}{2} = \frac{3\sqrt{2}h}{4}$$

Solution 3:



We have; $\theta = 1.22 \frac{\lambda}{a}$

Now, distance $= O_1O_2 = d\theta = 1.22 \frac{\lambda}{a} d$

And distance $= O_1O_2 = \frac{1.22 \times 5893 \times 10^{-10} \times 4 \times 10^8}{5} \approx 57.5 m \approx 60 m$

(Minimum distance)

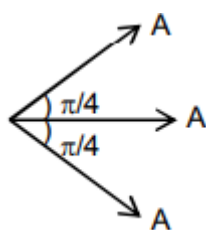
Solution 4:

Work: $W = \int \vec{F} \cdot d\vec{s}$

$$= \int_1^0 -x dx + \int_0^1 y dy$$

$$= -\frac{x^2}{2} \Big|_1^0 + \frac{y^2}{2} \Big|_0^1 = \left(0 + \frac{1}{2}\right) + \left(\frac{1}{2}\right) = 1J$$

Solution 5:



Resultant amplitude: $A_{res} = (\sqrt{2} + 1)A$

Resultant intensity: $I_{res} = (\sqrt{2} + 1)^2 I_0$

$= (3 + 2\sqrt{2}) I_0 = 5.8 I_{0s}$

Solution 6:

$$[f] = [ML^2T^{-2}]$$

Solution 7:

Using Parallel axes theorem

$$M.I \text{ about P} = 3 \left[\frac{2}{5} M \left(\frac{d}{2} \right)^2 + M \left(\frac{d}{\sqrt{3}} \right)^2 \right] = \frac{13}{10} M d^2$$

$$M.I \text{ about B} = 2 \left[\frac{2}{5} M \left(\frac{d}{2} \right)^2 + M (d)^2 \right] + \frac{2}{5} M \left(\frac{d}{2} \right)^2 = \frac{23}{10} M d^2$$

$$\text{Now ratio} = \frac{13}{23}$$

Solution 8:

Here $\vec{p} \cdot \vec{r} = 0$

$\vec{E}$ should be antiparallel to $\vec{p}$

$$\vec{E}, \vec{p}$$

$$\text{So, } \vec{E} = -\lambda(\vec{p})$$

where λ is a arbitrary positive constant

$$\text{Now } \vec{A} = a\hat{i} + b\hat{j} + c\hat{k}$$

$$\vec{A} = a\hat{i} + b\hat{j} + c\hat{k}$$

$$\vec{A} \parallel \vec{E}$$

$$\frac{a}{\lambda} = \frac{b}{3\lambda} = \frac{c}{-2\lambda} = k$$

$$\text{so } \vec{A} = \lambda k (\hat{i} + 3\hat{j} - 2\hat{k})$$

Solution 9:

Using Einstein's photoelectric equation;

$$KE_{\max} = E - \phi$$

$$= \frac{12400}{\lambda \left(\text{in } \overset{\circ}{\text{A}} \right)} - \phi \text{ (in eV)}$$

$$\therefore r = \frac{\sqrt{2mKE}}{eB}$$

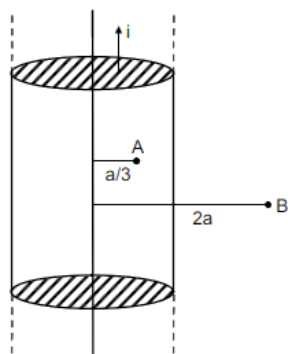
$$KE_{\max} = \frac{r^2 e^2 B^2}{2m} \text{ (in J)}$$

$$= \frac{r^2 e B^2}{2m} \text{ (in eV)}$$

$$\therefore \phi = \frac{12400}{6561} - \frac{r^2 e B^2}{2m} = 1.1 \text{ eV}$$

Solution 10:

Magnetic field at A:



$$B_A = \frac{\mu_0 i r}{2\pi a^2} = \frac{\mu_0 i \frac{a}{3}}{2\pi a^2} = \frac{\mu_0 i}{\pi a^2} \frac{a}{6} = \frac{\mu_0 i}{6\pi a}$$

Magnetic field at B;

$$B_B = \frac{\mu_0 i}{2\pi (2a)}$$

$$\frac{B_A}{B_B} = \frac{4}{6} = \frac{2}{3}$$

Solution 11:

We have;

$$E = \frac{\rho r}{3\epsilon_0}$$

$$E_A = \frac{-\rho R}{2(3\epsilon_0)}$$

$$|E_A| = \frac{\rho R}{6\epsilon_0}$$

$$E_B = E_{1A} + E_{2A}$$

$$E_{1A} = \text{Electric Field Due to solid sphere of radius R at point A} = \frac{\rho R}{3\epsilon_0}$$

E_{2A} = Electric Field Due to solid sphere of radius R/2 (which having charge density - ρ)

$$E_{2A} = R/2$$

$$= -\frac{KQ' \times 4}{9R^2} = -\frac{\rho R}{54\epsilon_0}$$

$$E_B = E_{1A} + E_{2A} = \frac{\rho R}{3\epsilon_0} - \frac{\rho R}{54\epsilon_0} = \frac{17\rho R}{54\epsilon_0}$$

$$\frac{|E_A|}{|E_B|} = \frac{9}{17}$$

Solution 12:

As per law of conservation of momentum

$$\frac{m}{2} \frac{v}{2} + mv = \left(m + \frac{m}{2}\right) v_f$$

$$v_f = \frac{5mV}{4 \times \frac{3m}{2}} = \frac{5V}{6}$$

$v_f < v_{orb} (=v)$ thus, the combined mass will take an elliptical path

Solution 13:

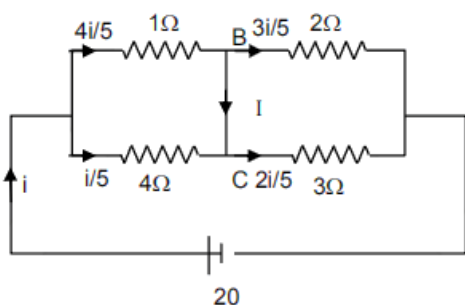
$$\text{Pitch of the screw} = \frac{3}{6} = 0.5 \text{ mm}$$

$$L.C. = \frac{0.5 \text{ mm}}{50} = \frac{1}{100} \text{ mm} = 0.01 \text{ mm}$$

$$= 0.001 \text{ cm}$$

Solution 14:

From the figure:



$$R_{eff} = \frac{4}{5} + \frac{6}{5} = 2\Omega$$

$$I = \frac{20}{2} = 10A$$

$$I = \frac{4i}{5} - \frac{3i}{5} = +\frac{i}{5} = 2A$$

Solution 15:

$$\vec{B}_1 = \frac{E_0}{c} \hat{k} \cos(kx - \omega t)$$

$$\vec{B}_2 = \frac{E_0}{c} \hat{i} \cos(ky - \omega t)$$

$$\vec{F} = q\vec{E} + q(\vec{V} \times \vec{B})$$

$$= q(\vec{E}_1 + \vec{E}_2) + q(\vec{V} \times \vec{B}_1 + \vec{B}_2)$$

by putting the values of $\vec{E}_1, \vec{E}_2, \vec{B}_1$ and $\vec{B}_2$

The net Lorentz force on the charged particle is

$$\vec{F} = qE_0 [0.8 \cos(kx - \omega t) \hat{i} + \cos(kx - \omega t) \hat{j} + 0.2 \cos(ky - \omega t) \hat{k}]$$

at $t = 0$ and at $x = y = 0$

$t = 0 \quad x = y = 0$

$$\vec{F} = qE_0 [0.8\hat{i} + \hat{j} + 0.2\hat{k}]$$

Solution 16:

$$(C_V)_A = \frac{5R}{2}$$

$$(C_V)_B = \frac{7R}{2}$$

$$\text{Now, } \frac{(C_V)_A}{(C_V)_B} = \frac{5}{7}$$

Solution 17:

Process A – B, isochoric

$PV = nRT$; as P increase T

Process B – C, adiabatic

$$PV^\gamma = \text{Constant}$$

$$\Rightarrow TV^{\gamma-1} = \text{Constant}$$

Process C – A, isobaric

$$V = kT$$

Solution 18:

Applying steady state equation of continuity

$$40 v_A = 20 v_B$$

$$\Rightarrow 2v_A = v_B$$

Using Bernoullies equation

$$P_A + \frac{1}{2} \rho v_A^2 = P_B + \frac{1}{2} \rho v_B^2$$

$$\Rightarrow P_A - P_B = \frac{1}{2} \rho (v_B^2 - v_A^2)$$

$$\Rightarrow \Delta P = \frac{1}{2} 1000 \left(v_B^2 - \frac{v_B^2}{4} \right)$$

$$\Rightarrow \Delta P = 500 \times \frac{3v_B^2}{4}$$

$$\Rightarrow v_B = \sqrt{\frac{(\Delta P) \times 4}{1500}} = \sqrt{\frac{(700) \times 4}{1500}} \text{ m/s}$$

$$\text{Volume flow rate} = 20 \times 100 \times v_B = 2732 \text{ cm}^3/\text{s}$$

Solution 19:

From law of conservation of momentum

$$mv\hat{i} + m\left(\frac{v}{2}\hat{i} + \frac{v}{2}\hat{j}\right) = 2m(v_1\hat{i} + v_2\hat{j})$$

$$\Rightarrow v_1 = \frac{3v}{4} \text{ and } v_2 = \frac{v}{4}$$

Change in K.E

$$\left[\frac{1}{2}mv^2 + \frac{1}{2}m\left(\frac{v}{2}\sqrt{2}\right)^2\right] - \left[\frac{1}{2}(2m)\left(\frac{9v^2}{16} + \frac{v^2}{16}\right)\right]$$

$$= \frac{3mv^2}{4} - \frac{5mv^2}{8} = \frac{mv^2}{8}$$

Solution 20:

de-Broglie equation;

$$\lambda = \frac{h}{\sqrt{2(KE)_m}} \Rightarrow \lambda \propto \frac{1}{\sqrt{KE}}$$

$$\frac{\lambda}{\lambda/2} = \sqrt{\frac{KE_f}{KE_i}}$$

$$4KE_i = KE_f$$

$$\Rightarrow \Delta E = 4KE_i - KE_i = 3KE = 3E$$

Solution 21:

Conservation of energy:

$$mg \frac{\ell}{2} \sin 30^\circ = \frac{1}{2} \frac{m\ell^2}{3} \omega^2$$

Then;

$$\omega^2 = 15$$

$$\omega = \sqrt{15}$$

Solution 22:

$$x^2 = at^2 + 2bt + c$$

Differentiating with respect to t ;

$$2xv = 2at + 2b$$

$$xv = at + b$$

$$v^2 + ax = a$$

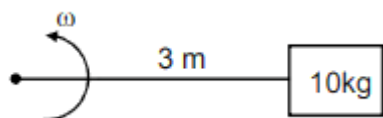
$$ax = a - \left(\frac{at+b}{x}\right)^2$$

$$a = \frac{a(at^2 + 2bt + c) - (at+b)^2}{x^3}$$

$$a = \frac{ac - b^2}{x^3}$$

$$a \propto x^{-3}$$

Solution 23:



Stress;

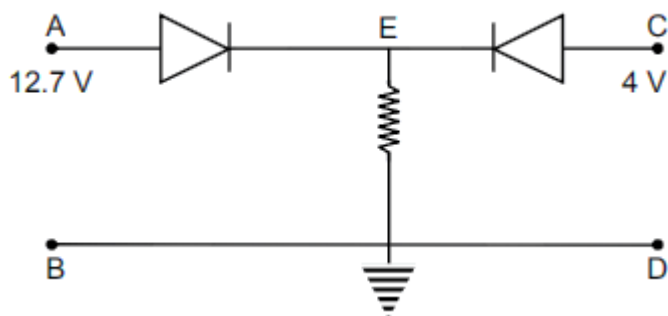
$$\frac{T}{A} = \sigma \quad \dots(1)$$

$$T = m\omega^2 \ell \quad \dots(2)$$

Solving

$$\omega = 4 \text{ rad / s}$$

Solution 24:



Right diode is reversed biased and left diode is forward biased

$$V_B = 0$$

$$\therefore V_E = 12.7 - 0.7 = 12 \text{ Volt}$$

Solution 25:

We have:

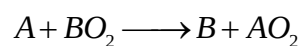
$$100 = \frac{L(0.25)}{0.025} \times 10^3$$

$$\therefore L = 100 \times 10^{-4} \text{ H}$$

$$= 10 \text{ mH}$$

CHEMISTRY

Solution 26:



$$\Delta G = -ve$$

Only above 1400°C

Solution 27:

Alkali metal oxides are basic in nature. Non-metal oxides are acidic in nature

Al_2O_3 is amphoteric. Amphoteric oxides are those which can behave both as acid as well as base.

Solution 28: When a molecule has unpaired electron then it will show magnetic moment.

$$O^2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2 \pi^* 2p_x^1 = \pi^* 2p_y^1$$

$$O_2^- = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2 \pi^* 2p_x^2 = \pi^* 2p_y^1$$

$$O_2^+ = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2 \pi^* 2p_x^1 = \pi^* 2p_y^0$$

Solution 29:

$$Q = [Pb^{2+}][Cl^-]^2$$

$$= \frac{300 \times 0.134}{400} \times \left[\frac{100 \times 0.4}{400} \right]^2$$

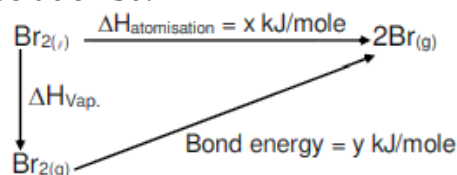
$$= \frac{3 \times 0.134}{4} \times (-0.1)^2$$

$$= 0.105 \times 10^{-2}$$

$$= 1.005 \times 10^{-3}$$

$$Q > K_{sp}$$

Solution 30:

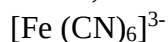


$$\Delta H_{\text{atomisation}} = \Delta H_{\text{vap}} + \text{Bond energy}$$

Hence $x > y$

Solution 31: Number of Geometrical isomers in square planar $[\text{PdFClBr}]^{2-}$ are = 3

Hence, $n = 3$



As cyanide ligand is strong ligands, it will result in pairing of electrons. $\text{Fe}^{3+} = 3d^5$,

According to CFT configuration is $t_{2g}^{221} e_g^{00}$

$$\mu = \sqrt{n(n+2)}$$

$$= 1.73 \text{ B.M.}$$

$$CFSE = -0.4\Delta_0 \times n_{t_{2g}} + 0.6\Delta_0 \times n_{e_g}$$

$$= -0.4 \Delta_0 \times 5 = -2.0\Delta_0$$

Solution 32:

$$2\pi r = n\lambda$$

$$2\pi \times \frac{n^2}{Z} a_0 = n\lambda$$

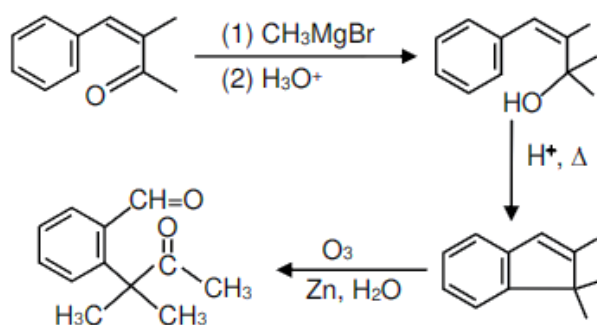
$$2\pi \times \frac{4^2}{1} a_0 = 4\lambda$$

$$\lambda = 8 \pi a_0$$

Solution 33: In H_3PO_4 Phosphorous is present its maximum oxidation number that is +5 state hence it cannot act as reducing agent as it can't get oxidized further .

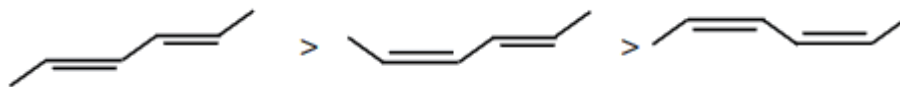
Solution 34: Conceptual.

Solution 35:



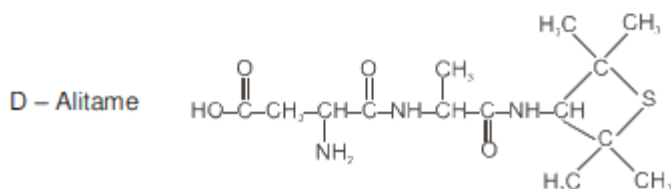
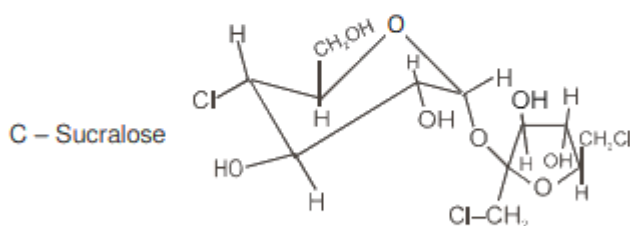
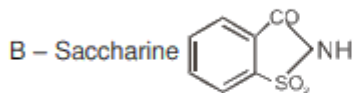
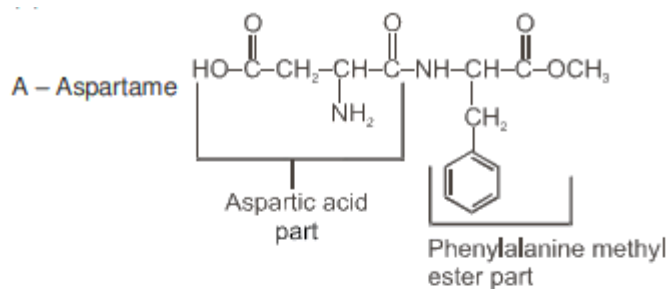
Solution 36: Aniline and phenol form complex with Lewis acid so most reactive among the given compounds for Friedel Craft reaction is chlorobenzene.

Solution 37: In isomers of hydrocarbon heat of combustion depends upon their stabilities. As the stability increases heat of combustion decreases.
Stability order



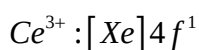
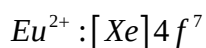
Solution 38: $\text{CCl}_4 \rightarrow$ Non-conductor in solid and liquid phase.

Solution 39:



- (i) A & D give positive test with ninhydrin because both have free carboxylic and amine groups.
- (ii) C form precipitate with AgNO_3 in the lassaigle extract of the sugar because it has chlorine atoms.
- (iii) B & D give positive test with sodium nitroprusside because both have sulphur atoms.

Solution 40:



Solution 41: $\text{Cr}(\text{H}_2\text{O})_6 \text{Cl}_n (\mu_{\text{complex}})_{\text{spin}} = 3.83 \text{ B.M.}$

From data of magnetic moment oxidation number of Cr should be +3

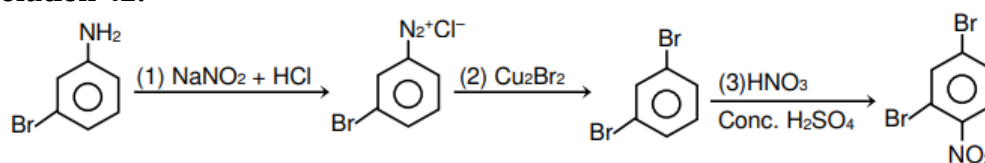
Hence complex is $\text{Cr}(\text{H}_2\text{O})_6 \text{Cl}_3$.

Complex shows geometrical isomerism therefore formula complex is $[\text{Cr}(\text{H}_2\text{O})_4 \text{Cl}_2] \text{Cl} \cdot 2\text{H}_2\text{O}$.

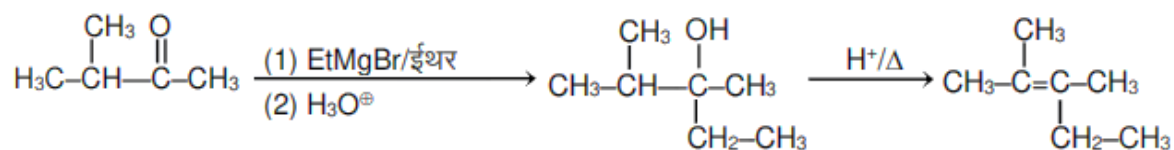
It's IUPAC Name : Tetraaquadichloridochromium (III) chloride dihydrate

$\text{Cr}(\text{H}_2\text{O})_6 \text{Cl}_n = 3.83 \text{ B.M.}$

Solution 42:



Solution 43:



Solution 44: Basicity is inversely proportional to electronegativity.

Solution 45:

$$K_{cat} = K$$

$$Ae^{-\frac{Ea_1}{RT_1}} = Ae^{-\frac{Ea_2}{RT_2}}$$

$$\frac{Ea_1}{T_1} = \frac{Ea_2}{T_2} \quad Ea_1 = Ea_2 - 30$$

$$\frac{Ea_2 - 30}{500} = \frac{Ea_2}{700}$$

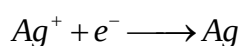
$$5Ea_2 = 7Ea_2 - 210$$

$$Ea_2 = \frac{210}{2} = 105 \text{ kJ / mole}$$

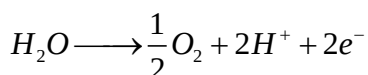
Activation energy of the catalysed reaction = $105 - 30 = 75 \text{ kJ/mole}$

Solution 46:

$$(n_{\text{Ag}})_{\text{deposit}} = \frac{108}{108} = 1 \text{ mole}$$



1F charge is required to deposit 1 mole of Ag



2F charge deposit $\longrightarrow \frac{1}{2}$ mole

1F charge will deposit $\longrightarrow \frac{1}{4}$ mole

$$V_{\text{O}_2} = \frac{nRT}{P}$$

$$= \frac{1}{4} \times \frac{0.08314 \times 273}{1}$$

$$V_{\text{O}_2} = 5.674 \text{ L}$$

Solution 47:

$$\Delta T_f = 0.2^\circ\text{C}$$

$$\Delta T_f = i k_f m$$

$$0.2 = 2 \times 2 \times \frac{w}{58.5} \times \frac{1000}{600}$$

$$w = \frac{0.2 \times 58.5 \times 600}{1000 \times 4}$$

$$= \frac{1.2 \times 58.5}{40}$$

$$= 1.76\text{g}$$

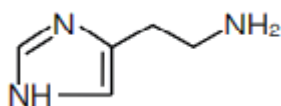
Solution 48: 10^{-3} molar $\text{MgSO}_4 = 10^{-3}$ moles of MgSO_4 present 1 L solutions

$$n_{\text{CaCO}_3} = n_{\text{MgSO}_4}$$

$$\text{ppm}_{(\text{in term of } \text{CaCO}_3)} = \frac{10^{-3} \times 100}{1000} \times 10^6$$

$$\text{ppm}_{(\text{in term of } \text{CaCO}_3)} = 100 \text{ ppm}$$

Solution 49: Structure of Histamine is



Molecular formula of Histamine is $\text{C}_5\text{H}_9\text{N}_3$

Molecular mass of Histamine is 111

$$\text{Percentage nitrogen by mass in Histamine} = \frac{42}{111} \times 100 = 37.84\%$$

Solution 50: 63% w/w $\rightarrow \text{HNO}_3$

$$M = \frac{63 \times 1.4}{63 \times 100} \times 1000 \text{ mole} / L = 14 \text{ mole} / L$$

MATHEMATICS

Solution 51:

$$\text{Given, } 2^{\frac{1}{4} + \frac{2}{16} + \frac{3}{48} + \dots \infty}$$

$$\Rightarrow 2^{\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \infty} = 2^{\frac{1}{2}} = \sqrt{2}$$

Solution 52:

$$\text{Given, } f'(x) = \tan^{-1}(\sec x + \tan x) = \tan^{-1}\left(\frac{1 + \sin x}{\cos x}\right)$$

$$= \tan^{-1} \left(\frac{1 - \cos \left(\frac{\pi}{2} + x \right)}{\sin \left(\frac{\pi}{2} + x \right)} \right)$$

$$= \tan^{-1} \left(\frac{2 \sin^2 \left(\frac{\pi}{4} + \frac{x}{2} \right)}{2 \sin \left(\frac{\pi}{4} + \frac{x}{2} \right) \cos \left(\frac{\pi}{4} + \frac{x}{2} \right)} \right)$$

$$= \tan^{-1} \left(\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right) = \frac{\pi}{4} + \frac{x}{2}$$

$$\int (f'(x)) dx = \int \left(\frac{\pi}{4} + \frac{x}{2} \right) dx$$

$$\Rightarrow f(x) = \frac{\pi}{4}x + \frac{x^2}{4} + c$$

$$\because f(0) = 0 = c$$

$$\Rightarrow f(x) = \frac{\pi}{4}x + \frac{x^2}{4}$$

$$\text{So, } f(1) = \frac{\pi+1}{4}$$

Solution 53:

$$\text{Let, } I = \int_0^{2\pi} \frac{x \sin^8 x}{\sin^8 x + \cos^8 x} dx \quad \dots\dots(i)$$

$$\text{Using property, } I = \int_0^{2\pi} \frac{(2\pi - x) \sin^8 x}{\sin^8 x + \cos^8 x} dx \quad \dots\dots(ii)$$

Adding (i) and (ii), we get

$$2I = \int_0^{2\pi} \left(\frac{x \sin^8 x}{\sin^8 x + \cos^8 x} + \frac{(2\pi - x) \sin^8 x}{\sin^8 x + \cos^8 x} \right) dx = \int_0^{2\pi} \frac{2\pi \sin^8 x}{\sin^8 x + \cos^8 x} dx$$

$$\Rightarrow I = 4\pi \int_0^{\pi/2} \frac{\sin^8 x}{\sin^8 x + \cos^8 x} dx$$

$$\Rightarrow 2I = 4\pi \int_0^{\pi/2} 1 dx$$

$$\Rightarrow I = 2\pi \times \frac{\pi}{2} = \pi^2$$

Solution 54:

$$\begin{aligned}
 \text{Given, } & \cos^3\left(\frac{\pi}{8}\right) \cdot \cos\left(\frac{3\pi}{8}\right) + \sin^3\left(\frac{\pi}{8}\right) \cdot \sin\left(\frac{3\pi}{8}\right) \\
 &= \cos^3 \frac{\pi}{8} \left[4\cos^3 \frac{\pi}{8} - 3\cos \frac{\pi}{8} \right] + \sin^3 \frac{\pi}{8} \left[3\sin \frac{\pi}{8} - 4\sin^3 \frac{\pi}{8} \right] \\
 &= 4\cos^6 \frac{\pi}{8} - 4\sin^6 \frac{\pi}{8} - 3\cos^4 \frac{\pi}{8} + 3\sin^4 \frac{\pi}{8} \\
 &= 4 \left[\left(\cos^2 \frac{\pi}{8} - \sin^2 \frac{\pi}{8} \right) \right] \left[\left(\sin^4 \frac{\pi}{8} + \cos^4 \frac{\pi}{8} + \sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8} \right) \right] - 3 \left[\left(\cos^2 \frac{\pi}{8} - \sin^2 \frac{\pi}{8} \right) \right] \\
 &= \cos \frac{\pi}{4} \left[4 \left(1 - \sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8} \right) - 3 \right] \\
 &= \frac{1}{\sqrt{2}} \left[1 - \frac{1}{2} \right] \\
 &= \frac{1}{2\sqrt{2}}
 \end{aligned}$$

Solution 55:

$$\text{Given, } A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{bmatrix}$$

$$\therefore |A| = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{vmatrix} = ((9+4) - 1(3-4) + 2(-1-3)) = 13+1-8 = 6$$

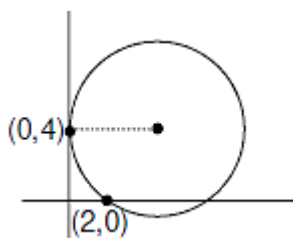
$$\therefore |adj B| = |adj adj A| = |A|^{(n-1)^2} = |A|^4$$

$$\Rightarrow |adj B| = (36)^2$$

$$\text{Also, } |C| = |3A| = 3^3 \times 6$$

$$\therefore \frac{|adj B|}{|C|} = \frac{36 \times 36}{3^3 \times 6} = 8$$

Solution 56:



Equation of family of circle will be

$$(x-0)^2 + (y-4)^2 + \lambda x = 0$$

Given it passes through (2, 0)

$$\therefore 4 + 16 + 2\lambda = 0$$

$$\Rightarrow \lambda = -10$$

$$\Rightarrow x^2 + y^2 - 10x - 8y + 16 = 0$$

Centre of above circle is (5, 4)

$$\Rightarrow \text{Radius } r = \sqrt{25 + 16 - 16} = 5$$

Thus, option (4) is correct as

$$\left| \frac{4 \times 5 + 3 \times 4 - 8}{5} \right| = \frac{24}{5} \neq 5$$

Solution 57:

Let thickness of ice = x cm

$$\therefore \text{Total volume } V = \frac{4}{3}\pi(10+x)^3$$

$$\frac{dV}{dt} = 4\pi(10+x)^2 \frac{dx}{dt} \quad \dots(i)$$

$$\text{Given } \frac{dV}{dt} = 50 \text{ cm}^3 / \text{min}$$

At $x = 5$ cm

$$50 = 4\pi(10+5)^2 \frac{dx}{dt}$$

$$\frac{dx}{dt} = \frac{1}{18\pi} \text{ cm/min}$$

Solution 58:

Let $p : \sqrt{5}$ is an integer

$q : 5$ is irrational

Thus, $\sim(p \vee q) = \sim p \wedge \sim q$

$\Rightarrow \sqrt{5}$ is not an integer and 5 is not an irrational number

Solution 59:

$$\text{Given, } I = \int \frac{dx}{(x+4)^{8/7} (x-3)^{6/7}}$$

$$\Rightarrow I = \int \left(\frac{x-3}{x+4} \right)^{-\frac{6}{7}} \frac{1}{(x+4)^2} dx$$

$$\text{Let } \frac{x-3}{x+4} = t^7$$

$$\Rightarrow \frac{7}{(x+4)^2} dx = 7t^6 dt$$

$$\Rightarrow I = \int t^{-6} t^6 dt = t + C$$

$$\Rightarrow I = \left(\frac{x-3}{x+4} \right)^{1/7} + C$$

Solution 60:

This can be done as by keeping 2 at 10th place as

			2	
--	--	--	---	--

$\therefore$ Number of such distinct numbers = $8 \times 8 \times 7 \times 6 = 2688 = 336k$

$$\Rightarrow k = 8$$

Solution 61:

The probability that the second A-card appears before the third B-card is

$$= AA + ABA + BAA + ABBA + BBAA + BABA$$

$$= \frac{1}{4} + \frac{1}{8} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16} = \frac{11}{16}$$

Solution 62:

Given, $\left| \frac{z-1}{z+2i} \right| = 1$ and $|z| = \frac{5}{2}$.

Let $z = x + iy$

$$\therefore x^2 + (y-1)^2 = x^2 + (y+2)^2$$

$$-2y + 1 = 4y + 4$$

$$6y = -3$$

$$\Rightarrow y = -\frac{1}{2}$$

Also, $x^2 + y^2 = \frac{25}{4}$

$$\Rightarrow x^2 = \frac{24}{4} = 6$$

$$\Rightarrow z = \pm\sqrt{6} - \frac{i}{2}$$

$$|z + 3i| = \sqrt{6 + \frac{25}{4}} = \sqrt{\frac{49}{4}}$$

$$\Rightarrow |z + 3i| = \frac{7}{2}$$

Solution 63:

For the intersection of the following three planes

$$x + 4y - 2z = 1$$

$$x + 7y - 5z = \beta$$

$$x + 5y + z = 5$$

$$\Delta = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 4 & -2 \\ 1 & 7 & -5 \\ 1 & 5 & \alpha \end{vmatrix} = 0$$

$$(7\alpha + 25) - (4\alpha + 10) + (-20 + 14) = 0$$

$$3\alpha + 9 = 0$$

$$\Rightarrow \alpha = -3$$

$$\text{Also } D_z = 0 \Rightarrow \begin{vmatrix} 1 & 4 & 1 \\ 1 & 7 & \beta \\ 1 & 5 & 5 \end{vmatrix} = 0$$

$$1(25 - 5\beta) - (15) + 1(4\beta - 7) = 0$$

$$\beta = 13$$

Solution 64:

$$\text{Given, } e^{4x} + e^{3x} - 4e^{2x} + e^x + 1 = 0$$

$$\text{Let } e^x = t \in (0, \infty)$$

$$\Rightarrow t^4 + t^3 - 4t^2 + t + 1 = 0$$

$$\Rightarrow t^2 + t - 4 + \frac{1}{t} + \frac{1}{t^2} = 0$$

$$\Rightarrow \left(t^2 + \frac{1}{t^2}\right) + \left(t + \frac{1}{t}\right) - 4 = 0$$

$$\text{Let } t + \frac{1}{t} = \alpha$$

$$(\alpha^2 - 2) + \alpha - 4 = 0$$

$$\alpha^2 + \alpha - 6 = 0$$

$$\alpha^2 = -3, 2$$

$$\Rightarrow \alpha = 2$$

$$\Rightarrow e^x + e^{-x} = 2$$

$x = 0$ is only solution

Solution 65:

Applying LMVT for $x \in [a, c]$

$$\frac{f(c) - f(a)}{c - a} = f'(\alpha), \alpha \in (a, c)$$

Applying LMVT for $x \in [c, b]$

$$\frac{f(b) - f(c)}{b - c} = f'(\beta), \beta \in (c, b)$$

$\therefore f''(x) < 0 \Rightarrow f'(x)$ is decreasing

$$f'(\alpha) > f'(\beta)$$

$$\frac{f(c) - f(a)}{c - a} > \frac{f(b) - f(c)}{b - c}$$

$$\frac{f(c) - f(a)}{f(b) - f(c)} > \frac{c - a}{b - c} \quad (\because f(x) \text{ is increasing})$$

Solution 66:

Here,

$$\int_0^1 (a + bx + cx^2) dx = ax + \frac{bx^2}{2} + \frac{cx^3}{3} \Big|_0^1 = a + \frac{b}{2} + \frac{c}{3}$$

$$\text{Also, } f(1) = a + b + c$$

$$f(0) = a$$

$$f\left(\frac{1}{2}\right) = a + \frac{b}{2} + \frac{c}{4}$$

Now from the options, we get

$$\begin{aligned} \frac{1}{6} \left(f(1) + f(0) + 4f\left(\frac{1}{2}\right) \right) &= \frac{1}{6} \left(a + b + c + a + 4 \left(a + \frac{b}{2} + \frac{c}{4} \right) \right) \\ &= \frac{1}{6} (6a + 3b + 2c) = a + \frac{b}{2} + \frac{c}{3} \end{aligned}$$

Solution 67:

$$\text{Mean } (x_i - 5) = \frac{\sum_{i=1}^{10} (x_i - 5)}{10} = 1$$

$$\therefore \lambda = \{\text{Mean}(x_i - 5)\} + 2 = 3$$

$$\mu = \text{var}(x_i - 5) = \frac{\sum (x_i - 5)^2}{10} - \frac{\sum (x_i - 5)}{10} = 3$$

Solution 68:

$$\text{For } \frac{x^2}{18} + \frac{y^2}{4} = 1$$

$$e_1 = \sqrt{1 - \frac{4}{18}} = \sqrt{\frac{7}{9}} = \frac{\sqrt{7}}{3}$$

$$\text{For } \frac{x^2}{9} - \frac{y^2}{4} = 1$$

$$e_2 = \sqrt{1 + \frac{4}{9}} = \sqrt{\frac{13}{9}} = \frac{\sqrt{13}}{3}$$

$\therefore (e_1, e_2)$ is a point on the ellipse, $15x^2 + 3y^2 = k$

$$\therefore 15e_1^2 + 3e_2^2 = k$$

$$\Rightarrow k = 15\left(\frac{7}{9}\right) + 3\left(\frac{13}{9}\right)$$

$$\therefore k = 16$$

Solution 69:

Given vertices of a triangle $(3, -1)$, $(1, 3)$ and $(2, 4)$

$\therefore$ Centroid $C = (2, 2)$

Also, point of intersection P is $\left(-\frac{1}{5}, \frac{2}{5}\right)$

$\therefore$ Equation of line CP

$$8x - 11y + 6 = 0$$

Solution 70:

$$\text{Given, } f(x) = \begin{cases} \frac{\sin(a+2)x + \sin x}{x}, & x < 0 \\ b, & x = 0 \\ \frac{(x+3x^2)^{\frac{1}{3}} - x^{\frac{1}{3}}}{x^{\frac{4}{3}}}, & x > 0 \end{cases}$$

At $x = 0$

$$\text{LHL} = \lim_{x \rightarrow 0^-} \frac{\sin(a+2)x + \sin x}{x} = a + 3$$

$$f(0) = b$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} \frac{(x + 3x^2)^{\frac{1}{3}} - x^{\frac{1}{3}}}{x^{\frac{4}{3}}} = \lim_{h \rightarrow 0} \left(\frac{(1 + 3h)^{\frac{1}{3}} - 1}{h} \right) = 1$$

$$\therefore a = -2$$

$$b = 1$$

$$\therefore a + 2b = 0$$

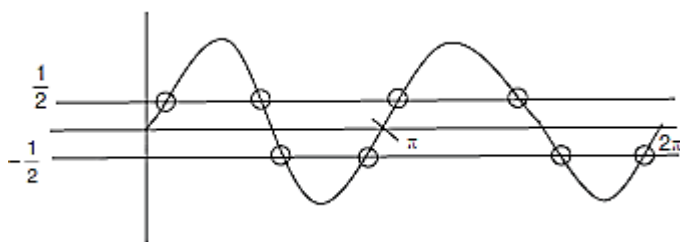
Solution 71:

$$\text{Given, } \log_{1/2} |\sin x| = 2 - \log_{1/2} |\cos x|$$

$$\Rightarrow \log_{1/2} |\sin x \cos x| = 2$$

$$\Rightarrow |\sin x \cos x| = \frac{1}{4}$$

$$\Rightarrow \sin 2x = \pm \frac{1}{2}$$



Clearly from the graph, number of solution = 8

Solution 72:

$$\text{Let } A(1, -1, 3), B(2, -4, 11), C(-1, 2, 3), D(3, -2, 10)$$

$$\therefore \overrightarrow{AB} = \hat{i} - 3\hat{j} + 8\hat{k} \text{ and } \overrightarrow{CD} = 4\hat{i} - 4\hat{j} + 7\hat{k}$$

$$\therefore \text{Projection of } \overrightarrow{AB} \text{ on } \overrightarrow{CD} = \frac{\overrightarrow{AB} \cdot \overrightarrow{CD}}{|\overrightarrow{CD}|} = \left(\frac{4 + 12 + 56}{\sqrt{16 + 16 + 49}} \right) = \frac{72}{9} = 8$$

Solution 73:

$$\text{Given, } (1 + x + x^2)^{10}$$

$$\therefore \text{General term } \frac{10!}{\alpha! \beta! \gamma!} 1^\alpha x^{\beta+2\gamma} \text{ such that } \alpha + \beta + 2\gamma = 10$$

Thus, for coefficient of $x^4 \Rightarrow \beta + 2\gamma = 4$

$$\gamma = 0, \beta = 4, \alpha = 6 \Rightarrow \frac{10!}{6!4!0!} = 210$$

$$\gamma = 1, \beta = 2, \alpha = 7 \Rightarrow \frac{10!}{7!2!1!} = 360$$

$$\gamma = 2, \beta = 0, \alpha = 8 \Rightarrow \frac{10!}{8!0!2!} = 45$$

$\therefore$ coefficient of $x^4 = 615$

Solution 74:

Given, $(x + 1)dy = \{(x + 1)^2 + y - 3\}dx$

$$\Rightarrow \frac{dy}{dx} = (1 + x) + \left(\frac{y - 3}{1 + x} \right)$$

$$\Rightarrow \frac{dy}{dx} - \frac{1}{(1 + x)}y = (1 + x) - \frac{3}{(1 + x)}$$

This is linear differential equation

$$\therefore I.F. = e^{-\int \frac{1}{(1+x)} dx} = \frac{1}{(1+x)}$$

$$\therefore \frac{d}{dx} \left(\frac{y}{1+x} \right) = 1 - \frac{3}{(1+x)^2}$$

$$\Rightarrow \frac{y}{1+x} = x + 3(1+x)^{-1} + c$$

$$\Rightarrow y = (1+x) \left[x + \frac{3}{(1+x)} + c \right]$$

$$\therefore \text{at } x = 2, y = 0$$

$$\therefore 0 = 3(2+1+c)$$

$$\Rightarrow c = -3$$

$$\therefore \text{at } x = 3, y = 3$$

Solution 75:

Given vectors, $\vec{p} = (a+1)\hat{i} + a\hat{j} + a\hat{k}$, $\vec{q} = a\hat{i} + (a+1)\hat{j} + a\hat{k}$ and $\vec{r} = a\hat{i} + a\hat{j} + (a+1)\hat{k}$

$\therefore$ For coplanar vectors

$$\begin{vmatrix} a+1 & a & a \\ a & a+1 & a \\ a & a & a+1 \end{vmatrix} = 0$$

$$\Rightarrow a+1+a+a=0$$

$$\Rightarrow a = -\frac{1}{3}$$

$$\vec{p} = \frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{1}{3}\hat{k}$$

$$\vec{q} = \frac{1}{3}(-\hat{i} + 2\hat{j} - \hat{k})$$

$$\vec{r} = \frac{1}{3}(-\hat{i} - \hat{j} + 2\hat{k})$$

$$\vec{p} \cdot \vec{q} = \frac{1}{9}(-2 - 2 + 1) = -\frac{1}{3}$$

$$\vec{r} \times \vec{q} = \frac{1}{9} \begin{vmatrix} i & j & k \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{vmatrix}$$

$$= \frac{1}{9}(i(4-1) - j(-2-1) + k(1+2))$$

$$= \frac{1}{9}(3i + 3j + 3k) = \frac{i + j + k}{3}$$

$$|\vec{r} \times \vec{q}| = \frac{1}{3}\sqrt{3} \Rightarrow |\vec{r} \times \vec{q}|^2 = \frac{1}{3}$$

$$3(\vec{p} \cdot \vec{q})^2 - \lambda(\vec{r} \times \vec{q})^2 = 0$$

$$3 \cdot \frac{1}{9} - \lambda \cdot \frac{1}{3} = 0$$

$$\Rightarrow \lambda = 1$$